

Turbojet Test and Analysis

Aerospace Propulsion

Laboratory Session 4



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1 Introduction

This laboratory session focuses on applying the analytical and computational tools developed throughout the course to a set of practical aerospace engineering problems. The data for this lab session was obtained from experimental measurements performed on the **Olympus HP turbojet** mounted on a dedicated test stand during a laboratory session.

The recorded variables allow the operating behaviour of the engine to be analysed under different running conditions, providing a direct link between measured quantities and the theoretical concepts discussed in class.

The aim is to strengthen the connection between theoretical models and real engineering behaviour. To this end, the experimental results are post-processed and interpreted, and later compared with numerical predictions obtained using **PROOSIS**, evaluating the engine performance at both **design** and **off-design** conditions. The translation of the data to **PROOSIS** makes it possible to assess the validity and limitations of the modelling assumptions and to identify discrepancies between idealised models and real engine operation.

The session is structured into several exercises, each targeting a different aspect of the methods introduced in class, ranging from data interpretation to performance analysis and model validation.

2 Evaluation of measurement Equipment calibration

2.1 Approximation of Specific Throttle Settings

In the first part we import the experimental data provided in a `.txt` file. The objective is to correctly read the dataset, extract the physical variables, and reorganise the information into **MATLAB** and `.xlsx` tables suitable for post-processing.

For this, we first plotted a **Tt4** graph to see the sections for **x** for which the turbojet finally stabilised after increasing the rotational speed of in **RPM**. After this, the x-axis spans for the stable regions are set and the mean value of the rest of the data was computed as seen on the tables.

In the figure below the stability regions are very clear, which is where we obtained the approximate data.

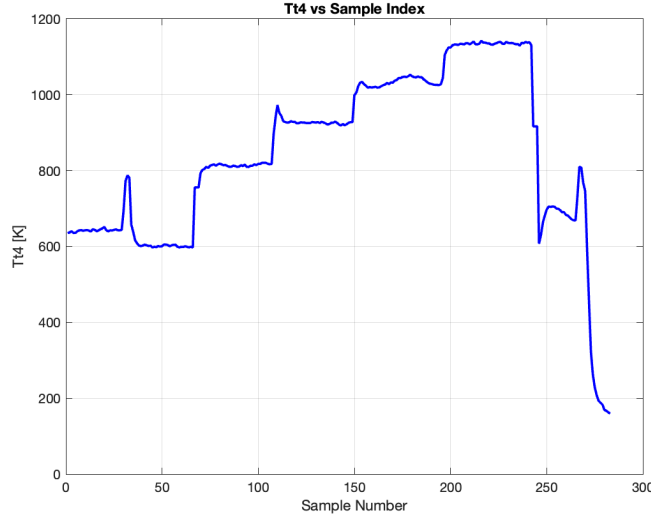


Figure 1: Tt4 vs x

N [rpm]	P_0 [bar]	T_0 [K]	P_{t2} [bar]	T_{t2} [K]	P_3 [bar]	P_{t3} [bar]	T_{t3} [K]	P_{t4} [bar]	T_{t4} [K]	P_{t5} [bar]	T_{t5} [K]	T_{t6} [K]
36000	0.931	286.475	0.931	286.457	0.222	0.181	310.810	0.175	915.090	0.035	771.240	740.550
50000	0.919	288.960	0.919	288.960	0.762	0.784	350.900	0.715	879.370	0.132	795.150	742.570
64800	0.903	288.214	0.903	288.214	1.449	1.551	399.190	1.389	1087.690	0.266	791.050	768.790
79200	0.895	288.280	0.895	288.280	1.895	1.994	429.490	1.829	1198.780	0.402	839.050	829.580
93600	0.891	288.310	0.891	288.310	2.212	2.301	450.500	2.143	1307.720	0.487	919.670	908.820
108000	0.890	288.310	0.890	288.310	2.495	2.574	470.890	2.424	1409.430	0.568	1021.130	1005.890

Table 1: Pressures and temperatures for different rotational speeds.

N [rpm]	Thrust [N]	$\dot{m}_f$ [kg/s]	$\dot{m}_a$ [kg/s]
36000	17.180	1.640×10^{-3}	0.031
50000	63.150	3.710×10^{-3}	0.127
64800	131.140	5.493×10^{-3}	0.243
79200	181.080	7.162×10^{-3}	0.310
93600	215.720	8.677×10^{-3}	0.350
108000	248.630	1.042×10^{-2}	0.381

Table 2: Thrust and mass flow rates for different rotational speeds.

Firstly, we decided that $P_0 \approx P_{t2}$ and $T_0 \approx T_{t2}$. This decision was made due to the fact that $M_0 = 0$, and under the assumption that the losses in the **HP Olympus Turbojet** are negligible.

From $N = 64800$ rpm onward, the compressor acts accordingly, which can be seen by the increase of P_{t3} and T_{t3} with N . That monotonic rise is exactly what you expect: higher spool speed $\Rightarrow$ higher pressure ratio and higher temperature rise across the compressor, however this is not the case everywhere.

It can be seen from **Table 1** that some of the pressure values do not look fully physical. For the lowest throttle settings (around 20% of the maximum RPM), the data shows $P_{t3} < P_{t2}$, which should not normally happen because the compressor should raise the total pressure. An initial estimate of this cause is stall, however this topic will be delved into more deeply further on in the report. Since turbine and compressor are mechanically coupled, failures in one parts function shows up in the other.

To check this, we plotted the time evolution of the pressures, as shown in the figure below. The steps in the signals match the throttle increments, and the compressor failure to compress only appears at the lowest speeds, which supports the idea that the engine is not operating in a stable compression regime there.

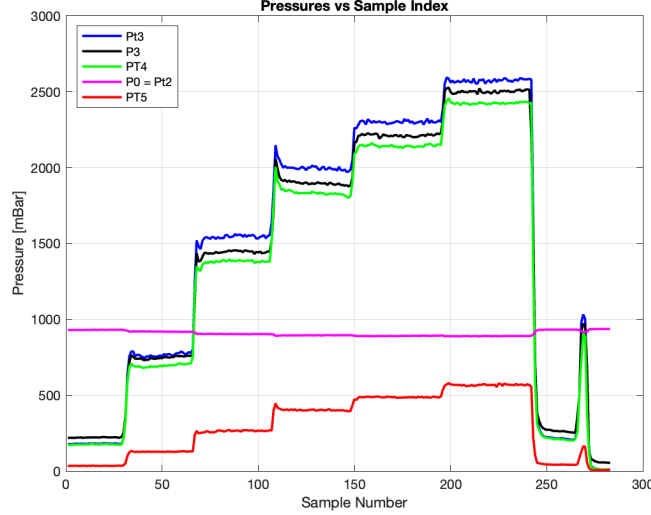


Figure 2: Measured pressures vs x

It is also observed that $P_3 < P_{t3}$ at most points, which is actually normal: P_3 is the *static* pressure after the compressor, while P_{t3} is the *total* pressure. On the contrary, from 0 to 20% the value of the static pressure is larger than the total one. As this should not happen, we assume stall is produced while the rotation velocity is at it's lower points. It is possible that this happens as a consequence of the power needed by the turbine to rotate being larger than the one produced by the whole engine.

On the other hand, thrust rises with speed from 17 N to 249 N. Air mass flow $\dot{m}_a$ also increases a lot ($0.031 \rightarrow 0.381$ kg/s), which is exactly why thrust rises so strongly: the engine is basically processing more air per second as the compressor pumps harder. Fuel-to-air ratio $f = \dot{m}_f / \dot{m}_a$ is on the order of a few percent: it drops from $\sim 5\%$ at 36k to $\sim 2-3\%$ for the higher-speed points, which is believable for small turbojets.

Also, the specific thrust $F / \dot{m}_a$ increases at high N , meaning the engine is not only swallowing more air, it's also doing a better job at turning that airflow into jet momentum at the top end.

2.1.1 Plots of Our Data vs The Theoretical Graphs Provided

In this part we are asked to interpret the meaning of each recorded channel and compute any missing or derived quantities required for the analysis. This may include checking units, converting between engineering conventions, and validating the integrity of the measurements before using them in later steps. The resulting comparison appears to be quite similar in most cases.

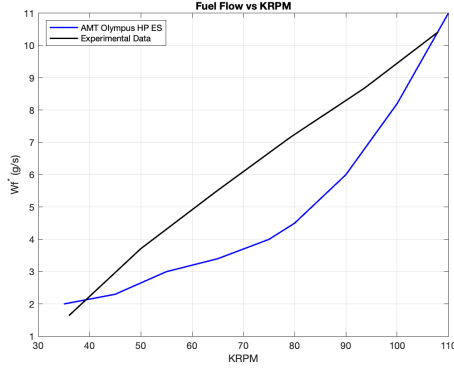


Figure 3: Fuel Flow vs Rotational Speed

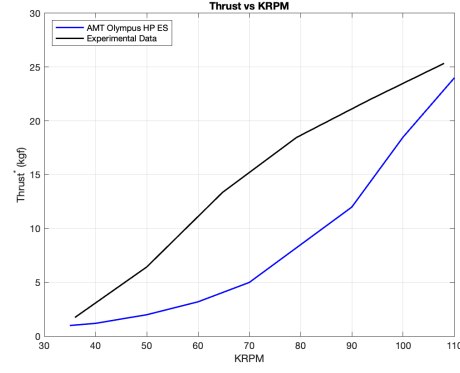


Figure 4: Thrust vs Rotational Speed

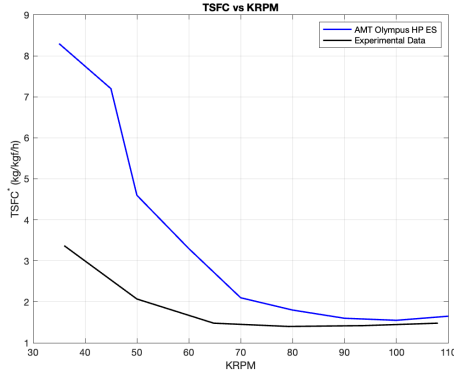


Figure 5: Specific Fuel Consumption vs Rotational Speed

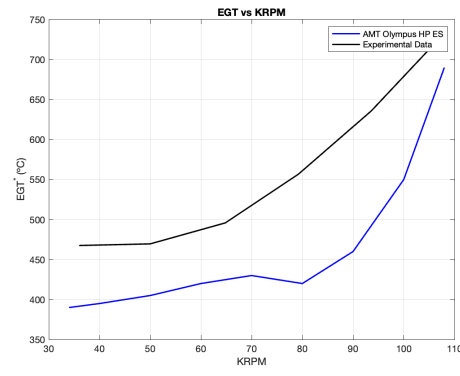


Figure 6: Exit Temperature T_6 vs Rotational Speed

It is observed that, though not the same, our data has similar tendencies as the one from the given tables. Specially in **Figures 3 and 4** the beginning and ending values match closely the ones from the **AMT Olympus HP ES**.

The **fuel mass flow rate** predicted by the model follows the experimental trend closely but in a more linear way, which makes sense. This indicates that the fuel scheduling implemented in the model is reasonably accurate near the design operating region, where combustion is more stable and control strategies are less influential.

The experimental **thrust** is higher than the model prediction for all rotational speeds, although both curves exhibit a similar increasing trend with KRPM. This indicates that the model correctly captures the qualitative relationship between rotational speed and thrust generation, but underestimates the absolute thrust level. Possible sources of this discrepancy include conservative assumptions regarding nozzle efficiency, air mass flow rate, or exhaust velocity.

The **Thrust Specific Fuel Consumption** (TSFC) is significantly higher in the model predictions at low rotational speeds, mainly due to the combined effect of overpredicted fuel flow and underpredicted thrust. As rotational speed increases, TSFC decreases for both the experimental data and the model, and the discrepancy between them is reduced. This confirms that the model is conservative in terms of efficiency, especially when operating far from the design point.

The experimental **Exhaust Gas Temperature** (EGT) is consistently higher than the model prediction over the entire operating range. This discrepancy is more pronounced at low and intermediate

rotational speeds and becomes smaller as the engine approaches maximum speed. This behaviour suggests that the model underestimates thermal losses and non-ideal combustion effects.

N [rpm]	Measured Thrust [kgf]	Specified Thrust [kgf]	Error [%]	Measured $\dot{m}_f$ [kg/s]	Specified $\dot{m}_f$ [kg/s]	Error [%]
36000	1.751	1.3	35	1.64×10^{-3}	2.07×10^{-3}	26.2
50000	6.437	2.5	157	3.71×10^{-3}	2.90×10^{-3}	28
64800	13.368	5.1	162	5.49×10^{-3}	3.74×10^{-3}	47
79200	18.460	9.3	98	7.16×10^{-3}	4.63×10^{-3}	54.6
93600	22.000	16	37.5	8.67×10^{-3}	6.72×10^{-3}	29
108000	25.345	23.1	9.7	1.042×10^{-2}	1.070×10^{-2}	2.7

Table 3: Measured and specified thrust and fuel mass flow.

N [rpm]	Measured TSFC [kg/kgf/h]	Specified TSFC [kg/kgf/h]	Error [%]	Measured EGT [°C]	Specified EGT [°C]	Error [%]
36000	3.37	8	137	467.55	395	18.4
50000	2.07	4.6	122	469.56	410	14.5
64800	1.48	2.8	89	495.79	425	16.7
79200	1.40	1.8	28.6	556.58	420	32.5
93600	1.42	1.55	9.2	635.82	490	29.8
108000	1.48	1.7	14.9	732.90	700	4.7

Table 4: Measured vs specified TSFC and EGT.

In terms of the values on table 3, and 4: As can be seen, all the measured parameters follow a similar trend to those specified, however, the rates at which these increase/decrease differ drastically, to the point that some of our values have an error of up to 162 percent. Overall, initial and final values tend to have low errors, meaning that they are very similar, the discrepancy arises from the other phases. During all experiments the measured engine outputs more thrust, consumes more fuel, and has a higher EGT. One possibility is that the turbojet used for measuring had higher efficiencies, meaning less losses, and thus allowing for higher values of exit velocities. Another possibility is that because the turbojet wasn't turned off completely, and was already partially warmed up from previous trials it had an easier time increasing its temperature and thus its velocity. Moreover, as can be seen in our properties chart, the compressor is not always compressing the air, on the contrary it at times decompresses it. Due to this happening at low RPM (as will be explained later on), these lower RPM parameters end up having unexpected values that could represent the high errors.

3 Component and Cycle Characterization

3.1 Calculation of Efficiencies

In this part, we used the previous calculations to compute the value of the efficiencies of the compressor η_c , turbine η_t , and the thermodynamic efficiency η_{th} .

Compressor Efficiency (η_c): The “Churning” Phase The compressor efficiency starts with negative values (-437% and -20%). This happens because at low RPMs (36k and 50k), the component is stalling rather than compressing. In a normal compressor, work is input to raise

N (rpm)	Compressor Efficiency (η_c)	Turbine Efficiency (η_t)	Thermal Efficiency (η_{th})
36000	-437.49 %	37.98 %	19.72 %
50000	-20.24 %	21.03 %	4.19 %
64800	41.56 %	77.08 %	7.55 %
79200	52.12 %	90.79 %	13.98 %
93600	53.60 %	92.51 %	18.06 %
108000	54.10 %	92.68 %	20.97 %

Table 5: Calculated Efficiencies based on Real vs. Isentropic Work

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pressure. However, because the test is static ($M_0 = 0$), the rotor creates high drag that lowers the pressure ($P_{t3} < P_{t2}$) while heating the air through friction ($T_{t3} > T_{t2}$). Since the efficiency equation compares the ideal work (negative here due to pressure drop) against real work input (positive temperature rise), the calculation results in a negative percentage. Once the RPM hits 64,800, the flow attaches, the pressure ratio exceeds 1.0, and efficiency becomes positive, stabilizing around 54%.

Turbine Efficiency (η_t): Turbine efficiency starts low ($\sim 21 - 38\%$) and rises to over 90%. This large variation is due to the turbine’s sensitivity to pressure ratio and incidence angle. At low RPM, the compressor does not provide enough pressure, so the pressure drop across the turbine is very low. Moreover, the airflow likely hits the blades at an off-design angle, causing separation rather than smooth rotation. The dip at 50,000 RPM likely represents a transition point where mass flow increases, but pressure is not yet high enough to sustain efficient expansion. As speed increases, the pressure builds and the flow angle corrects, allowing the efficiency to reach its higher values.

Thermal Efficiency (η_{th}): Pressure Ratio Dependence Thermal efficiency is extremely low at low speeds (4–7%) and improves to $\sim 21\%$. This is because the Brayton cycle efficiency depends directly on the overall Pressure Ratio. At 36k and 50k RPM, the compressor is not only not pressurizing the air, but actually depressurizing it, which is thermodynamically wasteful; fuel is burned to add heat, but very little net work is produced. As the RPM increases to 108,000, the pressure ratio increases to over 2.8, allowing the cycle to extract significantly more useful work per unit of fuel.

4 PROOSIS Implementation

To accurately simulate the performance of the micro-turbojet engine under static testing conditions, a partitioning strategy was implemented in PROOSIS using the variables summarized in the table below.

Unknown Variables	—
Boundary Conditions	$N_H, E_c, E_t, T_{t4}, Pl_c$
Algebraic Wizard	$f, \dot{m}_0$

Table 6: Partition for PROOSIS implementation.

The boundary conditions include the shaft rotational speed N_H , the compressor and turbine efficiencies (E_c and E_t), the turbine inlet total temperature T_{t4} , and the compressor pressure loss Pl_c . The

remaining variables, namely the fuel–air ratio f and the inlet mass flow rate $\dot{m}_0$, are solved algebraically by the model.

The engine schematic was modified to accurately represent static ground-test conditions. Since the engine is mounted on a stationary test bench rather than operating in flight, the inlet Mach number was set to $M_0 = 0$. This assumption eliminates any ram pressure rise, ensuring that the inlet conditions are purely dictated by the ambient atmosphere. The altitude was fixed to the average altitude of Madrid in order to define the corresponding ambient pressure and temperature.

A fixed nozzle exit area was also imposed as a boundary condition, with a value of $A_{\text{exit}} = 0.003137 \text{ m}^2$. This value corresponds to the geometry of the nozzle used in the experimental setup of the AMT Olympus micro-turbojet engine. As reported by S.M. Choi [1], the baseline circular conical nozzle employed in the experiments has an exit area of exactly 3137 mm^2 . Imposing this geometric constraint directly influences the mass flow rate and exit velocity calculations, ensuring consistency between the numerical model and the experimental configuration.

Following this setup, a standard reference case and six steady-state simulations were performed. Each operating point corresponds to a different rotational speed, with the boundary conditions and resulting thrust values reported in **Table 7**. For each case, the model parameters were carefully tuned to reproduce the experimentally measured thrust values, or values as close as possible to them. The selected parameter values were therefore chosen explicitly to satisfy this objective.

N (rpm)	η_c (%)	η_t (%)	p_{lc}	T_{t4} (K)	Thrust (kN)
36000	65	75	1.14	915.09	16.85
50000	65	82	1.4	879.37	64.012
64800	77	88	1.7176	1087.690	131.331
79200	75	90	2.228	1198.78	181.56
93600	65	92.51	2.5825	1307.72	215.59
108000	61	92.68	2.8921	1409.43	249.91

Table 7: Calculated Efficiencies and Performance Parameters

At the two lowest rotational speeds, the compressor pressure ratio was modified and the component efficiencies were significantly altered to replicate the experimental behavior. This adjustment was required because, under these operating conditions, the experimental compressor operates far from its nominal design point, effectively behaving in a manner opposite to its intended function. As a result, these operating points had to be reinterpreted within the PROOSIS framework.

For the remaining four operating points, the compressor and turbine efficiencies were progressively increased in order to achieve the target thrust values. Apart from these efficiency adjustments, no other parameters were modified.

References

- [1] S. M. Choi, S. Kim, R. S. Myong, and W. Kim, “Experimental investigation of infrared signal characteristics in a micro-turbojet engine,” *The Aeronautical Journal*, vol. 123, no. 1261, pp. 340–355, 2019.

Generative AI tools were used to assist in drafting and refining content, ensuring clarity and coherence. All outputs were reviewed and edited for accuracy and originality by the authors